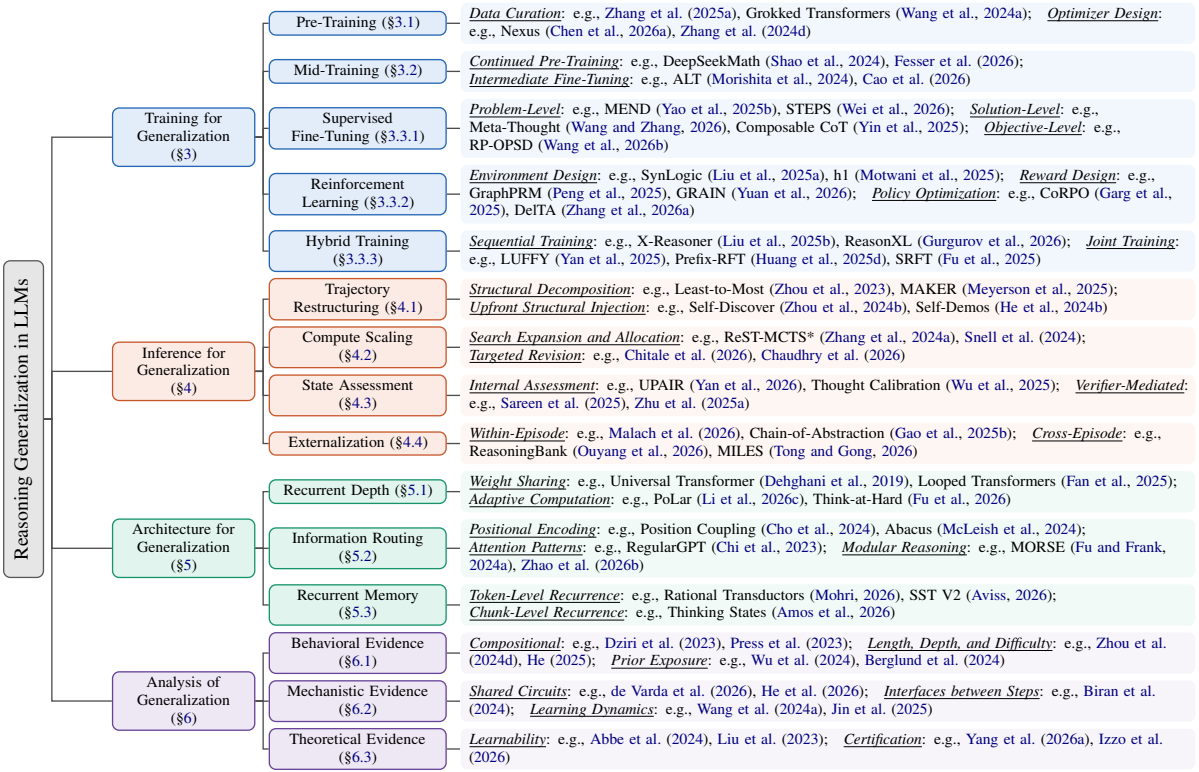

Reasoning Generalization in LLMs
Training for Generalization (§3)
Pre-Training (§3.1)
Data Curation: e.g., Zhang et al. (2025a), Grokked Transformers (Wang et al., 2024a); Optimizer Design: e.g., Nexus (Chen et al., 2026a), Zhang et al. (2024d)
Mid-Training (§3.2)
Continued Pre-Training: e.g., DeepSeekMath (Shao et al., 2024), Fesser et al. (2026); Intermediate Fine-Tuning: e.g., ALT (Morishita et al., 2024), Cao et al. (2026)
Supervised Fine-Tuning (§3.3.1)
Problem-Level: e.g., MEND (Yao et al., 2025b), STEPS (Wei et al., 2026); Solution-Level: e.g., Meta-Thought (Wang and Zhang, 2026), Composable CoT (Yin et al., 2025); Objective-Level: e.g., RP-OPSD (Wang et al., 2026b)
Reinforcement Learning (§3.3.2)
Environment Design: e.g., SynLogic (Liu et al., 2025a), h1 (Motwani et al., 2025); Reward Design: e.g., GraphPRM (Peng et al., 2025), GRAIN (Yuan et al., 2026); Policy Optimization: e.g., CoRPO (Garg et al., 2025), DelTA (Zhang et al., 2026a)
Hybrid Training (§3.3.3)
Sequential Training: e.g., X-Reasoner (Liu et al., 2025b), ReasonXL (Gurgurov et al., 2026); Joint Training: e.g., LUFFY (Yan et al., 2025), Prefix-RFT (Huang et al., 2025d), SRFT (Fu et al., 2025)
Inference for Generalization (§4)
Trajectory Restructuring (§4.1)
Structural Decomposition: e.g., Least-to-Most (Zhou et al., 2023), MAKER (Meyerson et al., 2025); Upfront Structural Injection: e.g., Self-Discover (Zhou et al., 2024b), Self-Demos (He et al., 2024b)
Compute Scaling (§4.2)
Search Expansion and Allocation: e.g., ReST-MCTS* (Zhang et al., 2024a), Snell et al. (2024); Targeted Revision: e.g., Chitale et al. (2026), Chaudhry et al. (2026)
State Assessment (§4.3)
Internal Assessment: e.g., UPAIR (Yan et al., 2026), Thought Calibration (Wu et al., 2025); Verifier-Mediated: e.g., Sareen et al. (2025), Zhu et al. (2025a)
Externalization (§4.4)
Within-Episode: e.g., Malach et al. (2026), Chain-of-Abstraction (Gao et al., 2025b); Cross-Episode: e.g., ReasoningBank (Ouyang et al., 2026), MILES (Tong and Gong, 2026)
Architecture for Generalization (§5)
Recurrent Depth (§5.1)
Weight Sharing: e.g., Universal Transformer (Dehghani et al., 2019), Looped Transformers (Fan et al., 2025); Adaptive Computation: e.g., PoLar (Li et al., 2026c), Think-at-Hard (Fu et al., 2026)
Information Routing (§5.2)
Positional Encoding: e.g., Position Coupling (Cho et al., 2024), Abacus (McLeish et al., 2024); Attention Patterns: e.g., RegularGPT (Chi et al., 2023); Modular Reasoning: e.g., MORSE (Fu and Frank, 2024a), Zhao et al. (2026b)
Recurrent Memory (§5.3)
Token-Level Recurrence: e.g., Rational Transductors (Mohri, 2026), SST V2 (Aviss, 2026); Chunk-Level Recurrence: e.g., Thinking States (Amos et al., 2026)
Analysis of Generalization (§6)
Behavioral Evidence (§6.1)
Compositional: e.g., Dziri et al. (2023), Press et al. (2023); Length, Depth, and Difficulty: e.g., Zhou et al. (2024d), He (2025); Prior Exposure: e.g., Wu et al. (2024), Berglund et al. (2024)
Mechanistic Evidence (§6.2)
Shared Circuits: e.g., de Varda et al. (2026), He et al. (2026); Interfaces between Steps: e.g., Biran et al. (2024); Learning Dynamics: e.g., Wang et al. (2024a), Jin et al. (2025)
Theoretical Evidence (§6.3)
Learnability: e.g., Abbe et al. (2024), Liu et al. (2023); Certification: e.g., Yang et al. (2026a), Izzo et al. (2026)